Grade 9 Mathematics

Exam study notes · November 2026
Paper 1 and Paper 2 (90 minutes, 75 marks each – bring a calculator, ruler, pencil and eraser)

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Paper 1

1. Algebraic expressions
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Paper 2

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1. Algebraic expressions

An **algebraic expression** uses letters (variables) to stand for numbers, e.g. **$3x^2 - 5x + 7$**. You must be able to name its parts, simplify it, multiply (expand) it and factorise it.

Words you must know

Word	Meaning	In $3x^2 - 5x + 7$
Term	Parts separated by + or − signs	3 terms: $3x^2$, $-5x$, 7
Coefficient	The number in front of a variable	3 (of x^2) and −5 (of x)
Constant	A term with no variable	7
Degree	The highest exponent	2
Monomial / binomial / trinomial	1 term / 2 terms / 3 terms	a trinomial
Like terms	Same variables with the same exponents	$3x^2$ and $-7x^2$ are like terms; $3x^2$ and $3x$ are not

Adding and subtracting: collect like terms

Only like terms can be added or subtracted. Add the coefficients and keep the variable part the same.

> $4a + 3b - a + 2b = (4a - a) + (3b + 2b) =$ **$3a + 5b$**
> $5x^2 - 2x + 3x^2 + 7x - 4 =$ **$8x^2 + 5x - 4$**

Multiplying

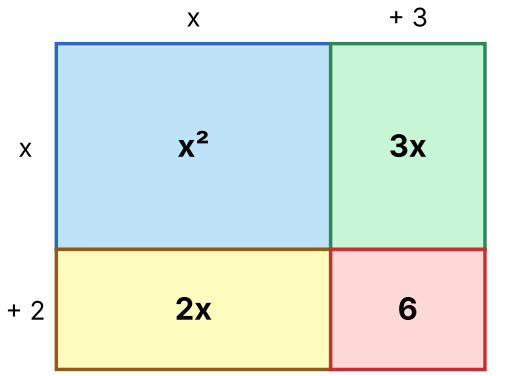
Multiply the **numbers** and add the **exponents** of the same variable: $3x^2y \times 4x^5y^3 = 12x^7y^4$.

Monomial × binomial: multiply every term inside the brackets

> $3x(2x - 5) = 3x \times 2x + 3x \times (-5) =$ **$6x^2 - 15x$**
> $-2a(a^2 + 4a - 1) =$ **$-2a^3 - 8a^2 + 2a$** (a minus outside changes every sign)

Binomial × binomial

Multiply **each term in the first bracket by each term in the second**. The area diagram shows why:



The four parts of the rectangle are the four products

$(x + 3)(x - 5) = x^2 - 5x + 3x - 15 = \mathbf{x^2 - 2x - 15}$
$(2x - 1)(x + 4) = 2x^2 + 8x - x - 4 = \mathbf{2x^2 + 7x - 4}$

Special products

Product	Rule	Example
Square of a sum	$(a + b)^2 = a^2 + 2ab + b^2$	$(x + 4)^2 = x^2 + 8x + 16$
Square of a difference	$(a - b)^2 = a^2 - 2ab + b^2$	$(2x - 3)^2 = 4x^2 - 12x + 9$
Difference of two squares	$(a + b)(a - b) = a^2 - b^2$	$(x + 7)(x - 7) = x^2 - 49$

Common mistake: $(x + 4)^2$ is **not** $x^2 + 16$. Write it as $(x + 4)(x + 4)$ and multiply out – the middle term $8x$ is easy to forget.

Dividing

Divide **every** term by the monomial: $(12x^3 - 8x^2 + 4x) \div 4x = \mathbf{3x^2 - 2x + 1}$.

Factorising (the opposite of expanding)

To **factorise** means to write an expression as a **product** (things multiplied). Always look for a **common factor first**.

Type	How	Examples
Common factor	Take out the highest number and variable that divide into every term	$6x^2 + 9x = 3x(2x + 3)$ $5a^2b - 10ab^2 = 5ab(a - 2b)$
Difference of two squares	$a^2 - b^2 = (a + b)(a - b)$. Both terms must be perfect squares with a minus between them	$x^2 - 25 = (x + 5)(x - 5)$ $9y^2 - 16 = (3y + 4)(3y - 4)$ $2x^2 - 50 = 2(x^2 - 25) = 2(x + 5)(x - 5)$
Trinomial $x^2 + bx + c$	Find two numbers that **multiply to c** and **add to b**	$x^2 + 7x + 12 = (x + 3)(x + 4)$ [3 × 4 = 12, 3 + 4 = 7] $x^2 - x - 12 = (x - 4)(x + 3)$ [−4 × 3 = −12, −4 + 3 = −1] $x^2 - 8x + 15 = (x - 3)(x - 5)$

> **Check:** you can always check a factorisation by multiplying it out again – you should get the original expression.

Simplifying algebraic fractions

Factorise the top and bottom first, then cancel **common factors** (never cancel single terms).

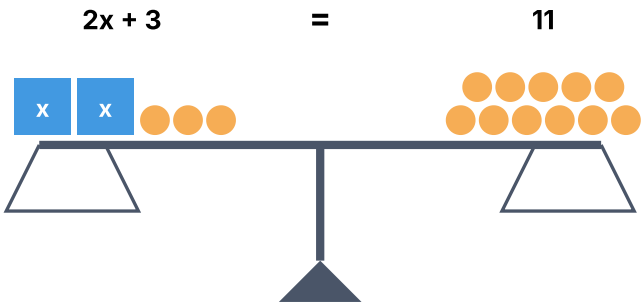
$(x^2 - 9) \div (x + 3) = (x + 3)(x - 3) \div (x + 3) =$ **$x - 3$**
$(4x + 8) \div 4 = 4(x + 2) \div 4 =$ **$x + 2$**

Substitution

Replace the variable with the number, using brackets: if $x = -2$, then $3x^2 - x = 3(-2)^2 - (-2) = 3(4) + 2 =$ **14**.

2. Algebraic equations

An **equation** is a number sentence with an equal sign. To **solve** it, find the value of the variable that makes both sides equal. An equation is like a **balanced scale**: whatever you do to one side, you must do to the other.



2x + 3 = 11: take 3 off both sides → 2x = 8; divide both sides by 2 → x = 4

Steps for solving

1. Remove **brackets** (multiply out).
2. Remove **fractions** by multiplying every term by the LCD (lowest common denominator).
3. Get all the **variable terms on one side** and the numbers on the other (do the opposite operation to both sides).
4. **Divide** by the coefficient of the variable.
5. **Check** by substituting your answer back into the original equation.

Type	Example	Solution
Simple linear	$3x - 7 = 11$	$3x = 18 \rightarrow$ **$x = 6$**
Variable on both sides	$5x + 2 = 2x - 10$	$3x = -12 \rightarrow$ **$x = -4$**
With brackets	$2(x - 3) = x + 4$	$2x - 6 = x + 4 \rightarrow$ **$x = 10$**
With fractions	$x/3 + x/2 = 5$	$\times 6$: $2x + 3x = 30 \rightarrow 5x = 30 \rightarrow$ **$x = 6$**
Product equals zero	$(x - 2)(x + 5) = 0$	$x - 2 = 0$ or $x + 5 = 0 \rightarrow$ **$x = 2$ or $x = -5$**
Factorise first	$x^2 - 5x + 6 = 0$	$(x - 2)(x - 3) = 0 \rightarrow$ **$x = 2$ or $x = 3$**
Common factor	$x^2 - 4x = 0$	$x(x - 4) = 0 \rightarrow$ **$x = 0$ or $x = 4$**
Exponential	$2^x = 32$	$2^x = 2^5 \rightarrow$ **$x = 5$**

Why "product = 0" works: if two numbers multiply to 0, at least one of them must be 0. So set **each bracket equal to 0**. Make sure one side is 0 **before** you factorise.

Checking: $5x + 2 = 2x - 10$ with $x = -4$: LHS $= 5(-4) + 2 = -18$; RHS $= 2(-4) - 10 = -18$. LHS = RHS ✓

Word problems

1. Let x = the unknown (write it down).
2. Turn the words into an equation.
3. Solve, then answer the question in words.

Example 1: The sum of three consecutive numbers is 72. Find the numbers.
Let the numbers be x, x + 1 and x + 2 → x + (x + 1) + (x + 2) = 72
3x + 3 = 72 → 3x = 69 → x = 23. The numbers are **23, 24 and 25**.

Example 2: Thabo is 3 times as old as his sister. In 4 years the sum of their ages will be 32. How old are they now?
Sister = x, Thabo = 3x → (x + 4) + (3x + 4) = 32
4x + 8 = 32 → 4x = 24 → x = 6. Sister is **6**, Thabo is **18**.

Example 3: The length of a rectangle is 5 cm more than its breadth. The perimeter is 38 cm. Find its dimensions.
Breadth = x, length = x + 5 → 2x + 2(x + 5) = 38
4x + 10 = 38 → 4x = 28 → x = 7. Breadth = **7 cm**, length = **12 cm**.

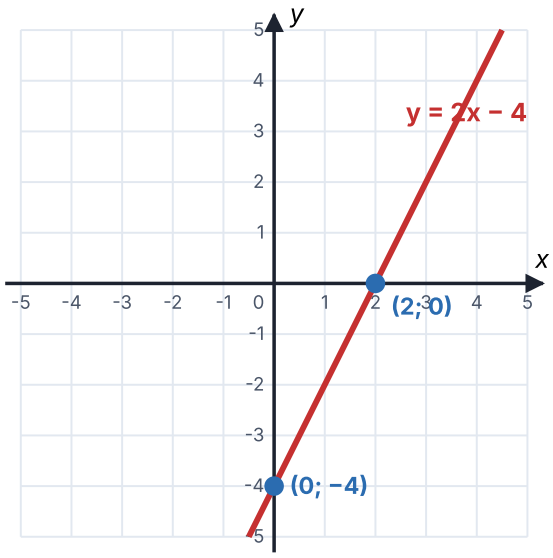
3. Graphs

A graph shows the relationship between two variables. On the **Cartesian plane**, the horizontal axis is the **x-axis** and the vertical axis is the **y-axis**. They cross at the **origin (0; 0)**. A point is written as **(x; y)** – x first, then y.

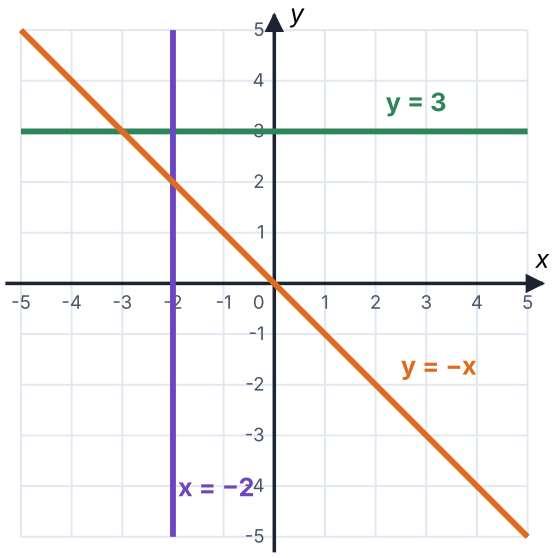
Linear graphs: y = mx + c

A **linear function** makes a **straight line**. In **y = mx + c**:

- **m = the gradient** (slope): how steep the line is. Gradient = rise ÷ run = $(y_2 - y_1) \div (x_2 - x_1)$.
  - m positive → the line goes **up** to the right; m negative → **down** to the right; m = 0 → **horizontal**.
- **c = the y-intercept**: where the line cuts the y-axis (put x = 0).
- The **x-intercept** is where the line cuts the x-axis (put y = 0 and solve).



y = 2x − 4: gradient 2, y-intercept −4, x-intercept 2



Special lines: horizontal, vertical, and through the origin

How to draw a straight-line graph

Method 1 – dual intercepts: draw y = 2x − 4
y-intercept: put x = 0 → y = −4 → point (0; −4)
x-intercept: put y = 0 → 0 = 2x − 4 → x = 2 → point (2; 0)
Plot both points and draw a straight line through them with a ruler. Label the line with its equation.

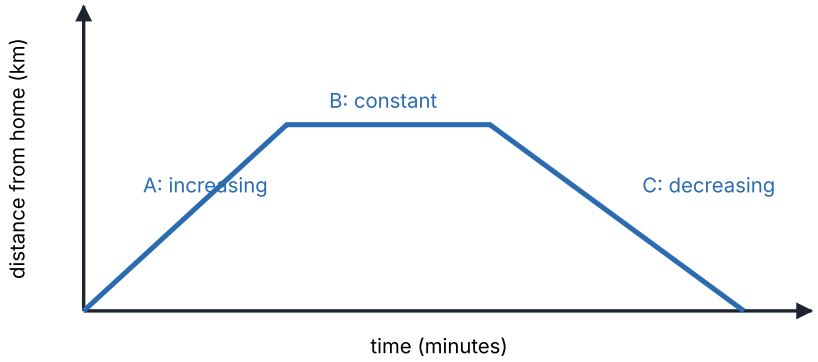
Method 2 – table of values: choose x-values, work out y, plot the points.
y = 2x − 4: x = −1 → y = −6; x = 0 → y = −4; x = 1 → y = −2; x = 2 → y = 0; x = 3 → y = 2

Gradient from two points: the line through (1; 3) and (4; 9)
m = (9 − 3) ÷ (4 − 1) = 6 ÷ 3 = **2**

Equation	What the line looks like
y = k (e.g. y = 3)	**Horizontal** line through 3 on the y-axis; gradient = 0
x = k (e.g. x = −2)	**Vertical** line through −2 on the x-axis; gradient is undefined
y = mx (no c)	Goes through the **origin** (0; 0)
Same m, e.g. y = 3x + 1 and y = 3x − 5	**Parallel** lines (same steepness)

Where two lines meet (point of intersection): y = x + 1 and y = −x + 5
Set them equal: x + 1 = −x + 5 → 2x = 4 → x = 2; then y = 2 + 1 = 3 → **(2; 3)**

Reading and interpreting graphs



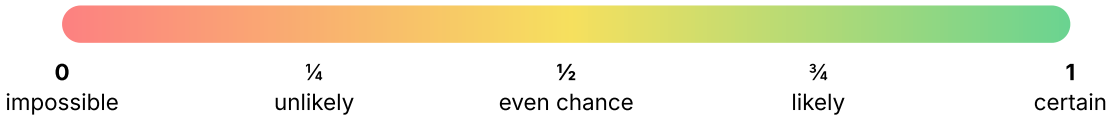
A learner walks away from home (A), stops at a shop (B) and walks back home (C)

Word	Meaning
Increasing	The graph goes up from left to right
Decreasing	The graph goes down from left to right
Constant	The graph is flat – no change (on a distance–time graph the person has stopped)
Linear / non-linear	A straight line / a curve
Discrete / continuous	Separate points that are not joined (e.g. number of people) / a joined line (e.g. time, temperature)
Maximum / minimum	The highest / lowest point of the graph

On a distance–time graph, a **steeper** line means **faster** movement.

4. Probability

Probability measures how likely something is to happen. It is a number from **0 (impossible)** to **1 (certain)**, written as a fraction, decimal or percentage.



The probability scale

P(event) = number of favourable outcomes ÷ total number of possible outcomes

Word	Meaning
Outcome	One possible result (rolling a 4)
Event	The outcome(s) you are interested in (rolling an even number)
Sample space	A list of **all** possible outcomes. Dice: {1; 2; 3; 4; 5; 6}
Complement	The event **not** happening: P(not A) = 1 − P(A)

One die:
P(even) = 3 ÷ 6 = ½ (2, 4, 6)
P(number greater than 4) = 2 ÷ 6 = ⅓ (5, 6)
P(not a 6) = 1 − ⅙ = ⅚ P(7) = **0** (impossible)

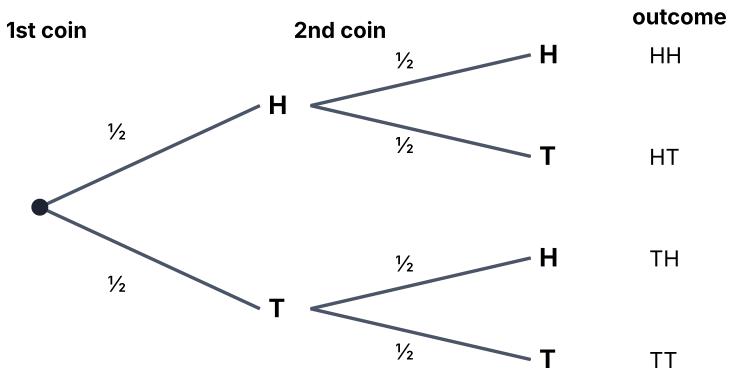
A bag has 5 red, 3 blue and 2 green beads (10 beads).
P(red) = 5 ÷ 10 = ½ P(blue) = **3/10** P(not green) = 8 ÷ 10 = ⅘

Two events: tables and tree diagrams

When two things happen together (e.g. a coin **and** a die), list all the outcomes in a **table** or a **tree diagram**. Total outcomes = outcomes of the first × outcomes of the second.

	1	2	3	4	5	6
H	H1	H2	H3	H4	H5	H6
T	T1	T2	T3	T4	T5	T6

Coin and die: 2 × 6 = 12 outcomes. P(heads and a 6) = 1/12. P(tails and an even number) = 3/12 = ¼.



Tree diagram for tossing two coins: 4 outcomes. P(two heads) = ¼; P(one head and one tail) = 2/4 = ½.

Theoretical probability and relative frequency

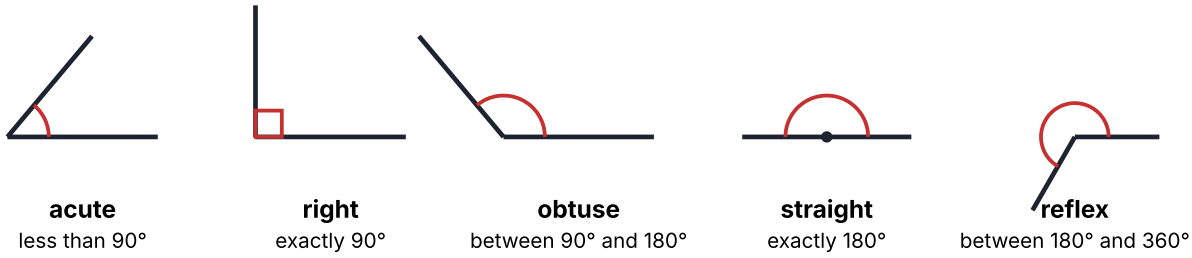
- **Theoretical probability:** what **should** happen, worked out with the formula. P(heads) = ½.
- **Relative frequency (experimental probability):** what **actually** happened in an experiment = number of times the event happened ÷ number of trials.
- The **more trials** you do, the **closer** the relative frequency gets to the theoretical probability.

Example: A coin is tossed 50 times and lands on heads 30 times.
Relative frequency of heads = 30 ÷ 50 = **0.6** (theoretical probability = 0.5)

5. Geometry of straight lines

In geometry you must give a **reason** for every statement, in brackets. Learn the facts **and** the short reasons.

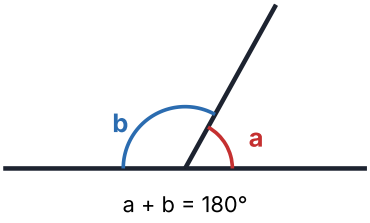
Types of angles



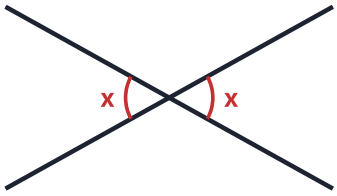
Types of angles

Also: a **revolution** is 360°. **Complementary** angles add up to **90°**; **supplementary** angles add up to **180°**.

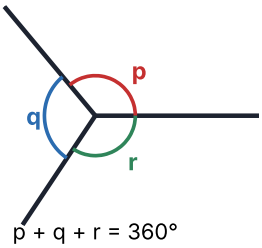
Angle facts on straight lines



∠s on a straight line add up to 180° (∠s on a str line)



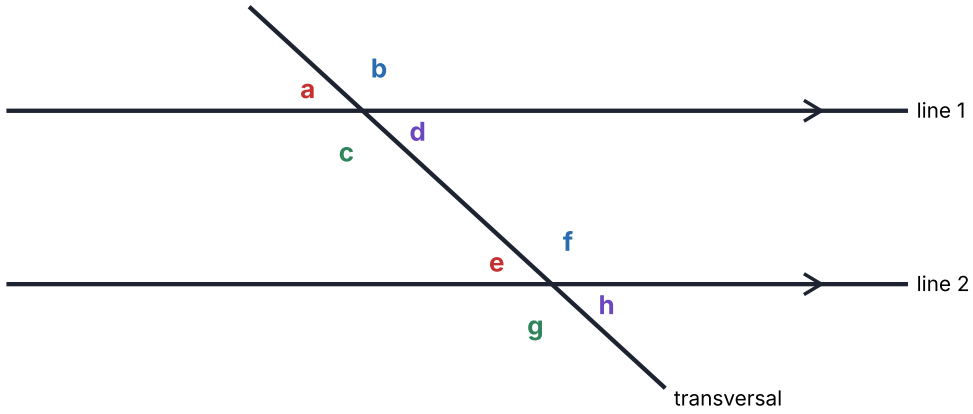
Vertically opposite ∠s are equal (vert opp ∠s)



∠s around a point add up to 360° (∠s round a pt)

Parallel lines cut by a transversal

A **transversal** is a line that cuts across two or more lines. When it cuts **parallel lines**, special pairs of angles are formed.



Line 1 ∥ line 2 (the arrows show the lines are parallel)

Angle pair	Shape to look for	Rule	In the diagram	Reason
Corresponding angles	**F**-shape (same position at each line)	Equal	a = e, b = f, c = g, d = h	(corr ∠s; line 1 ∥ line 2)
Alternate angles	**Z**-shape (inside the lines, opposite sides of the transversal)	Equal	c = f, d = e	(alt ∠s; line 1 ∥ line 2)
Co-interior angles	**U** / C-shape (inside, same side)	Add up to 180°	c + e = 180°, d + f = 180°	(co-int ∠s; line 1 ∥ line 2)

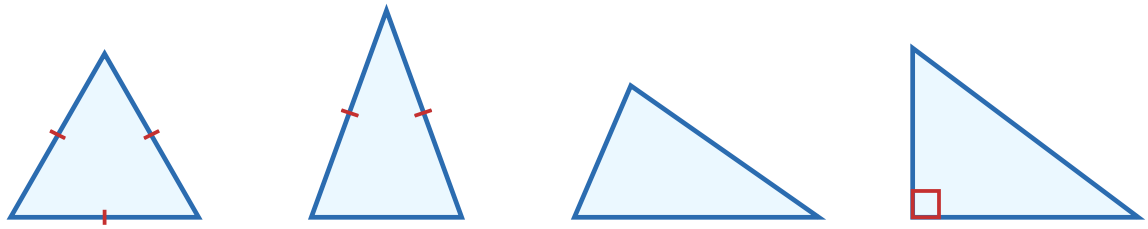
To prove lines are parallel, use the rules the other way round: if corresponding angles are equal, or alternate angles are equal, or co-interior angles add up to 180°, then the lines are parallel.

Example: In the diagram, a = 70°. Find all the other angles.
b = 180° − 70° = **110°** (∠s on a str line); d = a = **70°** (vert opp ∠s); c = b = **110°** (vert opp ∠s)
e = a = **70°** (corr ∠s); f = b = **110°** (corr ∠s); g = c = **110°**; h = d = **70°**

Other words: **perpendicular** lines meet at 90° (symbol ⊥); **parallel** lines never meet (symbol ∥); a **line segment** has two end points; a **ray** has one end point.

6. Geometry of 2D shapes: triangles and quadrilaterals

Types of triangles



Equilateral
3 equal sides, ∠s = 60°

Isosceles
2 equal sides and ∠s

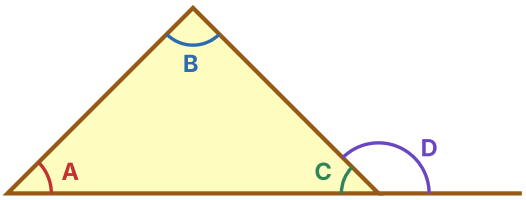
Scalene
no equal sides

Right-angled
one 90° angle

Tick marks show equal sides; a small square shows a right angle

Triangles can also be named by their angles: **acute-angled** (all angles less than 90°), **right-angled** (one 90°) and **obtuse-angled** (one angle more than 90°).

Angles of a triangle



A + B + C = 180° D = A + B

Interior angles and the exterior angle D

- The **interior angles** of a triangle add up to **180°** (sum of ∠s of △).
- The **exterior angle** of a triangle is equal to the **sum of the two opposite interior angles** (ext ∠ of △).
- In an **isosceles** triangle, the angles opposite the equal sides are equal (∠s opp equal sides).
- The longest side is opposite the biggest angle.

Example: A triangle has angles 2x, 3x and 85°. Find x.
2x + 3x + 85° = 180° (sum of ∠s of △) → 5x = 95° → **x = 19°**

Example: An isosceles triangle has a top angle of 40°. Find the two base angles.
The base angles are equal: (180° − 40°) ÷ 2 = **70°** each

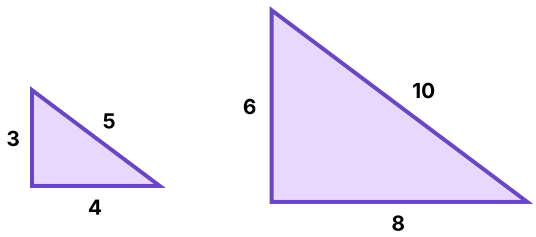
Congruent triangles (≡)

Congruent triangles are **exactly the same shape and size** – one fits perfectly on top of the other. Two triangles are congruent if one of these conditions is true:

Condition	What must be equal
SSS	All three sides
SAS	Two sides and the angle **between** them (included angle)
AAA	Two angles and a corresponding side
RHS	A right angle, the hypotenuse and one other side

AAA and SSA do **not** prove congruency. Write the letters in matching order: △ABC ≡ △DEF means A matches D, B matches E, C matches F.

Similar triangles (|||)



Every side of the big triangle is 2 times the small one

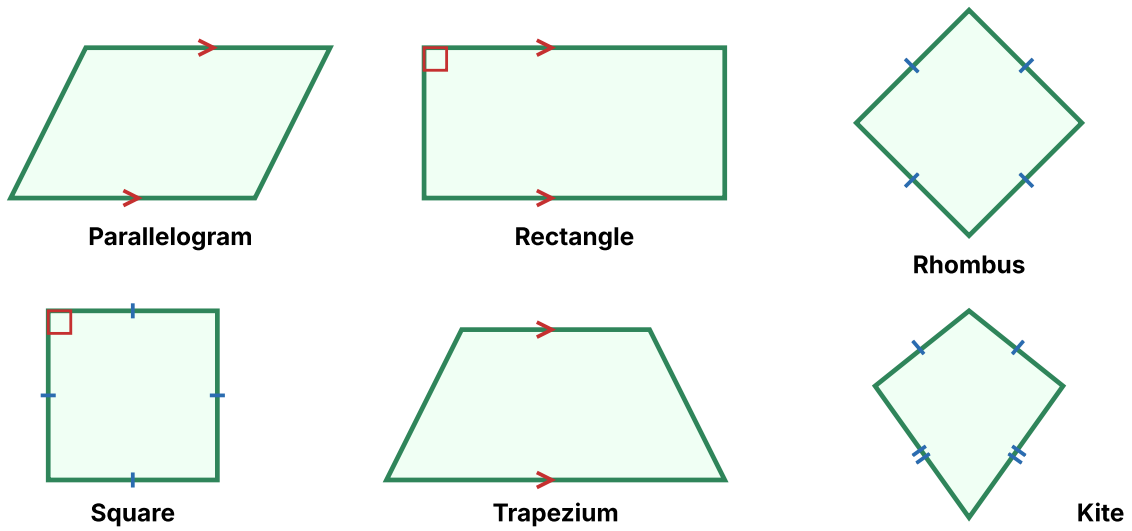
Similar triangles have the **same shape but a different size**. Two triangles are similar if:

- all their corresponding **angles are equal**, or
- all their corresponding **sides are in the same ratio** (proportion).

Example: △ABC ||| △DEF. AB = 4, DE = 8 and BC = 5. Find EF.
Ratio = DE ÷ AB = 8 ÷ 4 = 2 → EF = 5 × 2 = **10**

Quadrilaterals

A **quadrilateral** has **four sides**. Its interior angles add up to **360°**.



Arrows show parallel sides; tick marks show equal sides

Shape	Sides	Angles	Diagonals
Parallelogram	Both pairs of opposite sides parallel and equal	Opposite angles equal	Bisect each other (cut each other in half)
Rectangle	Opposite sides parallel and equal	All four angles 90°	Equal in length and bisect each other
Rhombus	All four sides equal; opposite sides parallel	Opposite angles equal	Bisect each other at 90°; bisect the angles
Square	All four sides equal; opposite sides parallel	All four angles 90°	Equal, bisect each other at 90°, bisect the angles
Trapezium	**One** pair of opposite sides parallel	–	–
Kite	Two pairs of **adjacent** sides equal	One pair of opposite angles equal	Perpendicular; the long one bisects the short one

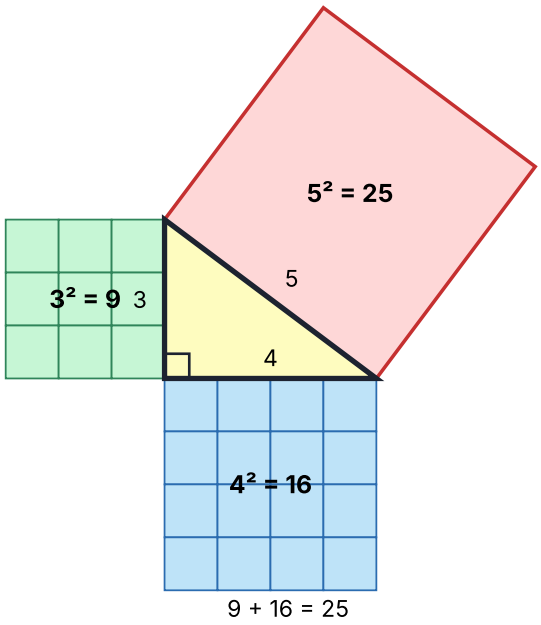
A square is a special rectangle **and** a special rhombus. A rectangle and a rhombus are special parallelograms.

Example: Three angles of a quadrilateral are 85°, 110° and 75°. Find the fourth angle.
360° − (85° + 110° + 75°) = 360° − 270° = **90°** (sum of ∠s of a quad)

The interior angles of **any polygon** add up to **180° × (n − 2)**, where n is the number of sides. A pentagon: 180° × 3 = 540°; a hexagon: 180° × 4 = 720°.

7. The Theorem of Pythagoras

The Theorem of Pythagoras only works in a **right-angled triangle**. The longest side, opposite the right angle, is called the **hypotenuse**.



The square on the hypotenuse = the sum of the squares on the other two sides

> **Theorem of Pythagoras:** in a right-angled triangle, the square on the hypotenuse is equal to the sum of the squares on the other two sides.
>
> **$c^2 = a^2 + b^2$**
> (c = hypotenuse)

Use it to:

- find the **hypotenuse** (add the squares),
- find a **shorter side** (subtract the squares),
- **test** whether a triangle is right-angled.

Find the hypotenuse: the two shorter sides are 6 cm and 8 cm.
$c^2 = 6^2 + 8^2 = 36 + 64 = 100 \rightarrow c = \sqrt{100} =$ **10 cm**

Find a shorter side: the hypotenuse is 13 cm and one side is 5 cm.
$a^2 = 13^2 - 5^2 = 169 - 25 = 144 \rightarrow a = \sqrt{144} =$ **12 cm**

Is it right-angled? A triangle has sides 7, 24 and 25.
Longest side$^2 = 25^2 = 625$. Other two: $7^2 + 24^2 = 49 + 576 = 625$. They are equal, so **yes, it is right-angled** (converse of Pythagoras).

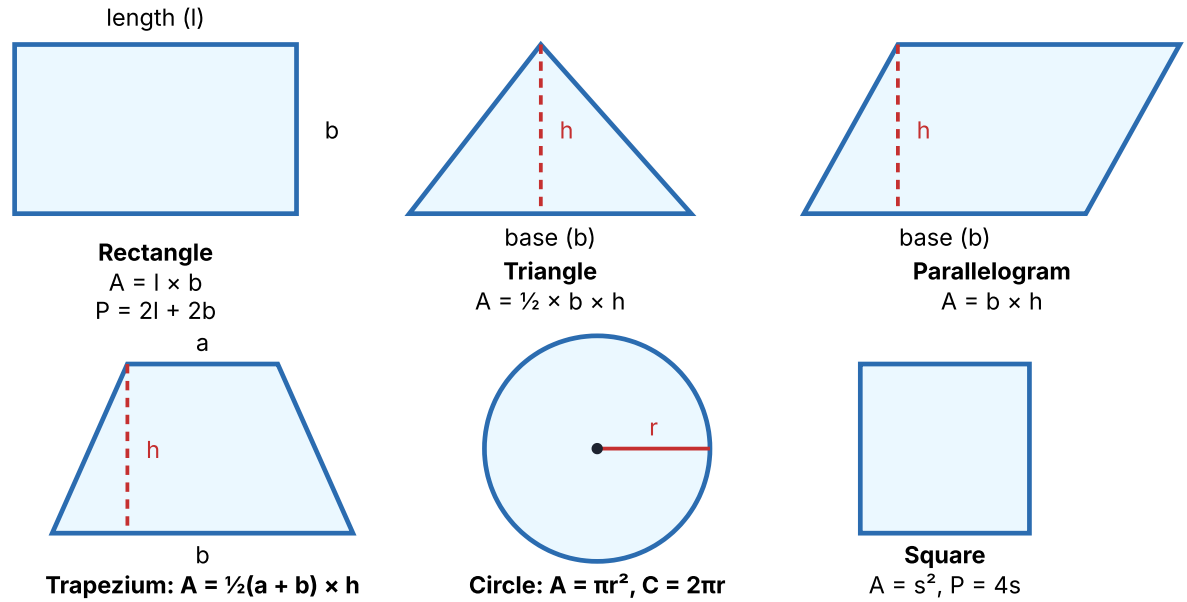
Word problem: a 5 m ladder leans against a wall. Its foot is 3 m from the wall. How high up the wall does it reach?
The ladder is the hypotenuse: $h^2 = 5^2 - 3^2 = 25 - 9 = 16 \rightarrow h =$ **4 m**

Answer that is not a whole number: sides 4 cm and 7 cm.
$c^2 = 16 + 49 = 65 \rightarrow c = \sqrt{65} =$ **8.06 cm** (round to 2 decimal places)

Pythagorean triples (whole-number right triangles): 3-4-5, 5-12-13, 8-15-17, 7-24-25 and their multiples (6-8-10, 9-12-15 …).

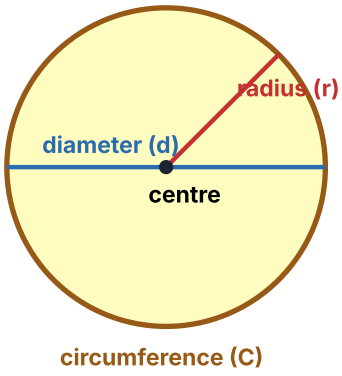
8. Perimeter and area

Perimeter is the **distance around** a shape (units: mm, cm, m). **Area** is the **space inside** a flat shape (units: mm², cm², m²).



*Area formulas. The height (h, red dashes) is always **perpendicular** (at 90°) to the base.*

The circle



Parts of a circle

- **Radius (r):** from the centre to the edge.
- **Diameter (d):** right across through the centre. **d = 2r**.
- **Circumference (C):** the perimeter of a circle. **C = 2πr** (or **πd**).
- **Area: A = πr²**.
- **π ≈ 3.14** (or use the π key on your calculator).

Circle example: a circle has a diameter of 10 cm.
r = 10 ÷ 2 = 5 cm
Circumference: C = 2 × 3.14 × 5 = **31.4 cm**
Area: A = 3.14 × 5² = 3.14 × 25 = **78.5 cm²**

Triangle example: base 12 cm, perpendicular height 7 cm. A = ½ × 12 × 7 = **42 cm²**

Trapezium example: parallel sides 6 m and 10 m, height 4 m. A = ½ × (6 + 10) × 4 = ½ × 16 × 4 = **32 m²**

Combined shape: a rectangle 10 m × 6 m with a semicircle of diameter 6 m on one end.
Rectangle: 10 × 6 = 60 m²; semicircle: ½ × 3.14 × 3² = 14.13 m²; total = **74.13 m²**

Find a missing side: a rectangle has an area of 48 cm² and a length of 8 cm. Breadth = 48 ÷ 8 = **6 cm**

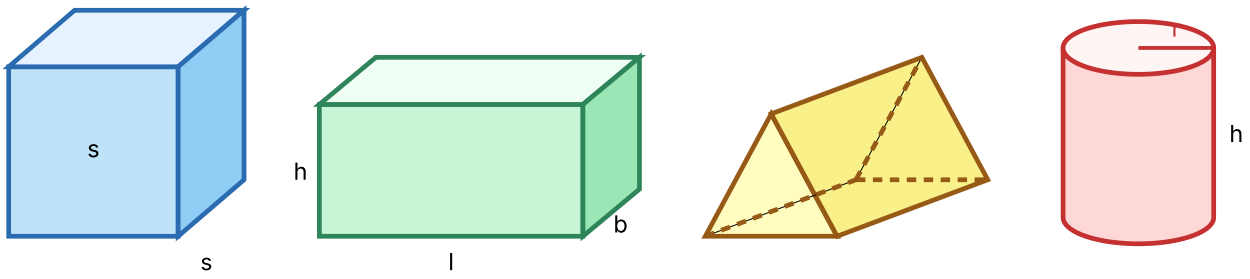
Converting units

Length	Area (square the conversion)
1 cm = 10 mm	1 cm² = 10 × 10 = **100 mm²**
1 m = 100 cm	1 m² = 100 × 100 = **10 000 cm²**
1 km = 1 000 m	1 km² = 1 000 000 m²

If all the lengths of a shape are multiplied by k, the **perimeter** is multiplied by k and the **area** by **k²**. Doubling the sides of a square makes its area 4 times bigger.

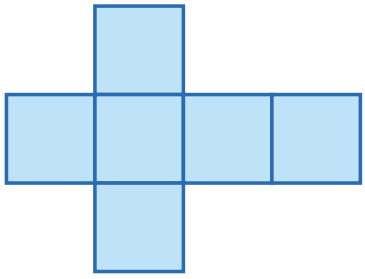
9. Volume and surface area

Surface area is the **total area of all the faces** of a 3D object (units: cm², m²). **Volume** is the **space inside** a 3D object (units: cm³, m³).



The 3D objects you must know

Nets and surface area



6 equal squares → SA = 6s²

A net of a cube

A **net** is the flat shape you get when you unfold a 3D object. The surface area is the **area of the net**: add the areas of all the faces.

Object	Surface area
Cube	$6s^2$
Rectangular prism	$2(lb + lh + bh)$
Triangular prism	2 × area of triangle + area of the 3 rectangles
Cylinder	$2\pi r^2 + 2\pi rh$ (two circles + the curved side)

Volume

Volume of any prism = area of the base × height (the base is the shape that is the same all the way through).

Object	Volume
Cube	$s \times s \times s = s^3$
Rectangular prism	$l \times b \times h$
Triangular prism	(½ × base × height of the triangle) × length of the prism
Cylinder	$\pi r^2 \times h$

Rectangular prism 5 cm × 3 cm × 4 cm:
Volume = 5 × 3 × 4 = **60 cm³**
Surface area = 2(5×3 + 5×4 + 3×4) = 2(15 + 20 + 12) = 2 × 47 = **94 cm²**

Cube with sides of 4 cm: Volume = 4^3 = **64 cm³**; Surface area = 6 × 4^2 = 6 × 16 = **96 cm²**

Cylinder r = 7 cm, h = 10 cm (π = 3.14):
Volume = 3.14 × 7^2 × 10 = 3.14 × 49 × 10 = **1 538.6 cm³**
Surface area = 2 × 3.14 × 7^2 + 2 × 3.14 × 7 × 10 = 307.72 + 439.6 = **747.32 cm²**

Triangular prism: the triangle has a base of 6 cm and a height of 4 cm; the prism is 10 cm long.
Volume = (½ × 6 × 4) × 10 = 12 × 10 = **120 cm³**

Volume and capacity

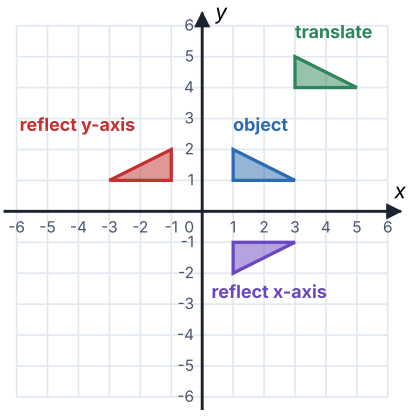
Volume	Capacity
1 cm^3	= 1 mℓ
1 000 cm^3	= 1 ℓ
1 m^3	= 1 000 ℓ = 1 kℓ

Example: a fish tank 50 cm × 30 cm × 40 cm holds 50 × 30 × 40 = 60 000 cm^3 = **60 ℓ** of water.

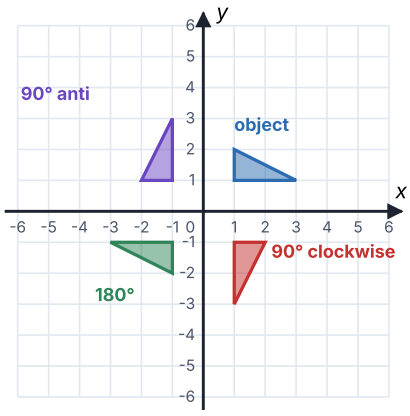
If all the lengths are multiplied by k, the surface area is multiplied by **k^2** and the volume by **k^3**. Doubling the sides of a cube makes its volume **8 times** bigger.

10. Transformations

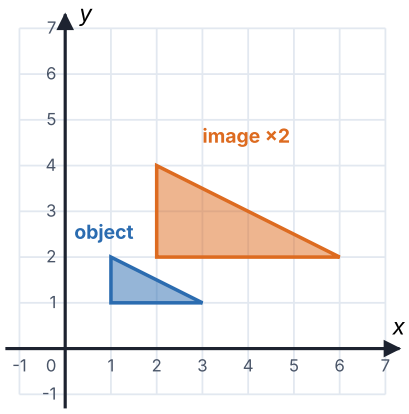
A **transformation** moves or changes a shape. The original shape is the **object**; the new shape is the **image**, labelled with a dash: A → A′.



Reflections and a translation (2 right, 3 up)



Rotations about the origin



Enlargement by a factor of 2 from the origin

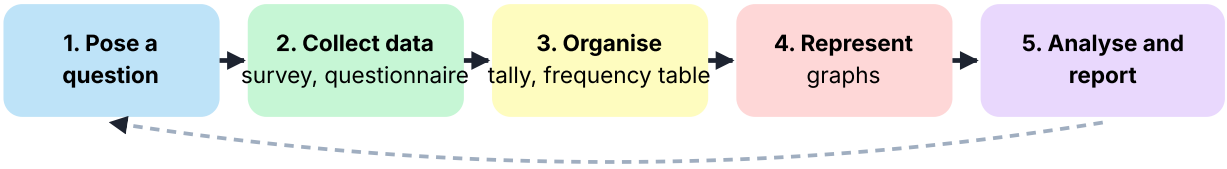
The blue **object** triangle has corners (1; 1), (3; 1) and (1; 2).

Transformation	What happens	Rule	Object (3; 1) becomes
Translation (slide)	Every point moves the same distance in the same direction	(x; y) → (x + a; y + b) 2 right, 3 up: (x + 2; y + 3)	(5; 4)
Reflection in the y-axis	Mirror image across the y-axis	(x; y) → (−x; y)	(−3; 1)
Reflection in the x-axis	Mirror image across the x-axis	(x; y) → (x; −y)	(3; −1)
Reflection in y = x	Swap x and y	(x; y) → (y; x)	(1; 3)
Rotation 90° clockwise about the origin	A quarter turn clockwise	(x; y) → (y; −x)	(1; −3)
Rotation 90° anticlockwise	A quarter turn anticlockwise	(x; y) → (−y; x)	(−1; 3)
Rotation 180°	A half turn	(x; y) → (−x; −y)	(−3; −1)
Enlargement by factor k from the origin	Shape gets bigger (k > 1) or smaller (0 < k < 1)	(x; y) → (kx; ky)	k = 2: (6; 2)

Same or different? Translations, reflections and rotations give an image that is **congruent** (same shape and size). An enlargement or reduction gives an image that is **similar** (same shape, different size). With a scale factor k, the area is multiplied by $\mathbf{k^2}$.

11. Data handling

Data handling is collecting, organising, showing and interpreting information (data) to answer a question.



The data cycle

Collecting data

- **Population:** the whole group you want to know about. **Sample:** a smaller part of the population that you actually ask.
- A sample must be **representative** (like the whole population) and **not biased** (not favouring one answer). Asking only rugby players about the favourite sport of the school would be biased.
- **Categorical data:** words or groups (favourite colour). **Numerical data:** numbers – **discrete** (counted, e.g. number of siblings) or **continuous** (measured, e.g. height, mass).

Organising data

Tally and frequency table – marks: 25 learners' test marks (out of 10)

Mark	Tally	Frequency
5	𝍸	4
6	𝍸 |	6
7	𝍸 |||	8
8	𝍸	5
9	||	2
Total		**25**

Stem-and-leaf display – the stem is the tens digit, the leaves are the units: data 23, 25, 31, 34, 34, 38, 42, 47

Stem	Leaf
2	3 5
3	1 4 4 8
4	2 7

Measures of central tendency and spread

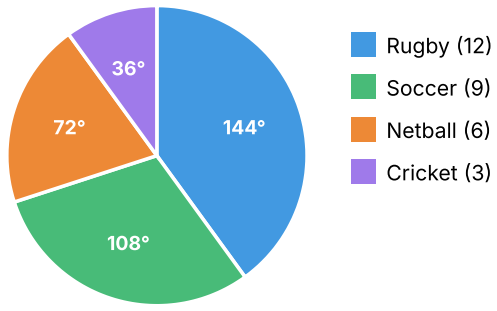
Data: **4; 7; 7; 9; 13**

Measure	What it is	How to find it	Answer
Mean (average)	The typical value	Add all values ÷ number of values	40 ÷ 5 = **8**
Median	The middle value	Put the data **in order** and find the middle. With an even number of values, add the two middle ones ÷ 2	**7**
Mode	The most common value	The value that appears most often (there can be more than one, or none)	**7**
Range	How spread out the data is	Highest value − lowest value	13 − 4 = **9**

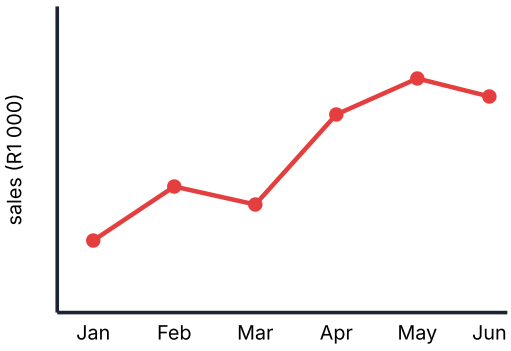
Median of an even set: 3; 5; 8; 10; 12; 15 → middle two are 8 and 10 → median = (8 + 10) ÷ 2 = **9**

An **outlier** is a value much bigger or smaller than the rest. It pulls the **mean** up or down a lot, so the **median** is a better "typical" value when there are outliers.

Graphs



Pie chart: favourite sport of 30 learners



Broken-line graph: change over time

Graph	Use it for	Important
Bar graph	Comparing categories (favourite sport, colours)	Bars are the same width with equal **gaps**
Double bar graph	Comparing two sets of data side by side (boys vs girls)	Use a key
Histogram	Grouped numerical data in intervals (marks 0–9, 10–19 ...)	Bars **touch** – no gaps
Pie chart	Parts of a whole (how a total is divided)	Angle of a sector = (frequency ÷ total) × 360°
Broken-line graph	Showing a change over time (temperature, sales)	Time goes on the horizontal axis
Scatter plot	Showing whether two variables are related (correlation)	Positive, negative or no correlation

Pie chart angles for 30 learners: Rugby 12 → (12 ÷ 30) × 360° = **144°**; Soccer 9 → **108°**; Netball 6 → **72°**; Cricket 3 → **36°**. Check: 144 + 108 + 72 + 36 = 360° ✓

Percentage from a pie chart: Rugby = 12 ÷ 30 × 100 = **40%** of the learners

Interpreting and reporting

- Describe what the graph shows: the **biggest** and **smallest** values, **trends** (going up or down), and any **outliers**.
- Use the mean, median, mode and range to support your conclusions.
- **Misleading graphs:** watch for a vertical axis that does not start at 0, uneven scales, missing labels or pictures of different sizes – these can make a small difference look big.

Definitions to learn by heart

These use the usual Grade 9 wording. If your teacher or textbook words a definition differently, learn their wording.

Term	Definition
Term	Parts of an expression separated by + or − signs.
Coefficient	The number in front of a variable.
Like terms	Terms with the same variables and the same exponents.
Factorise	Write an expression as a product of its factors.
Equation	A mathematical sentence with an equal sign; solving it means finding the value of the variable that makes it true.
Gradient	The steepness of a line: rise ÷ run.
y-intercept	The point where a graph cuts the y-axis ($x = 0$).
x-intercept	The point where a graph cuts the x-axis ($y = 0$).
Probability	How likely an event is to happen, from 0 (impossible) to 1 (certain).
Sample space	The list of all possible outcomes.
Relative frequency	The number of times an event happens ÷ the number of trials.
Acute / right / obtuse / straight / reflex angle	Less than 90° / 90° / between 90° and 180° / 180° / between 180° and 360°.
Complementary angles	Two angles that add up to 90°.
Supplementary angles	Two angles that add up to 180°.
Vertically opposite angles	The opposite angles formed when two lines cross; they are equal.
Transversal	A line that cuts across two or more other lines.
Corresponding angles	Angles in the same position at each line (F-shape); equal if the lines are parallel.
Alternate angles	Angles between the lines on opposite sides of the transversal (Z-shape); equal if the lines are parallel.
Co-interior angles	Angles between the lines on the same side of the transversal (U-shape); add up to 180° if the lines are parallel.
Perpendicular lines	Lines that meet at 90°.
Parallel lines	Lines that are always the same distance apart and never meet.
Congruent	Exactly the same shape and size.
Similar	The same shape but a different size; angles equal and sides in proportion.

Hypotenuse	The longest side of a right-angled triangle, opposite the right angle.
Theorem of Pythagoras	In a right-angled triangle, the square on the hypotenuse equals the sum of the squares on the other two sides.
Perimeter	The total distance around a shape.
Area	The amount of flat space inside a shape, in square units.
Circumference	The perimeter of a circle.
Radius / diameter	The distance from the centre to the edge of a circle / the distance across a circle through the centre (d = 2r).
Surface area	The total area of all the faces of a 3D object.
Volume	The amount of space inside a 3D object, in cubic units.
Net	A flat shape that folds up to make a 3D object.
Translation	Sliding a shape so every point moves the same distance in the same direction.
Reflection	Flipping a shape over a mirror line.
Rotation	Turning a shape around a fixed point.
Enlargement	Making a shape bigger or smaller by a scale factor.
Mean	The sum of all values ÷ the number of values.
Median	The middle value when the data is in order.
Mode	The value that occurs most often.
Range	The highest value − the lowest value.
Outlier	A value that is much bigger or smaller than the rest of the data.
Population / sample	The whole group being studied / a smaller part of it that is actually surveyed.